SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone Project Report & Presentation

Author:	Pandya Shashank (Pandya Nareshbhai)
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1. Executive Summary

Project Overview, Methodology & Key Results

Objective: Predict whether the SpaceX Falcon 9 first stage booster will land successfully to evaluate launch economics and determine competitive bidding strategies.

Phase	Methods Applied	Key Findings & Deliverables
Data Collection	SpaceX REST API v4 + Web Scraping Wikipedia table with BeautifulSoup	90 Falcon 9 launches collected (2010–2020), 17 core telemetry features
Data Wrangling & EDA	Landing outcome binary class labeling (Class 1/0), SQL aggregation, Seaborn charts	Overall landing success rate: **66.67%**. 80 encoded features. Strong upward yearly trend.
Visual Analytics	Folium interactive maps + Plotly Dash dashboard	Coastal launch site safety buffers; KSC LC-39A success rate 76.2%; 2,000–6,000 kg optimal payload
Predictive Modeling	Logistic Regression, SVM, Decision Tree, KNN tuned via GridSearchCV (10-fold CV)	**83.33% Test Accuracy** across all models; 100% recall (0 False Negatives). Best model: Decision Tree.

2. Introduction & Problem Statement

Economics of Booster Reusability in Commercial Spaceflight

Background: SpaceX launch price of ~ vs competitors + is driven by reusability of the first stage booster (representing ~70% of vehicle hardware cost).

- **Problem:** Predicting landing success enables commercial aerospace bidders to estimate SpaceX cost structures and bid competitively.

- **Research Questions:**

1. Which launch sites and orbits yield the highest booster recovery success rates?
2. How does payload mass impact recovery outcome across landing platforms (ASDS vs RTLS vs Ocean)?
3. How has landing success evolved across booster iterations from 2010 to 2020?
4. Which supervised ML classification model achieves the highest landing prediction accuracy?

3. Data Collection – SpaceX REST API

REST API Extraction & Entity Normalization

- **Endpoint:** https://api.spacexdata.com/v4/launches/past returning nested JSON records.
- **Normalization:** pd.json_normalize() flattened nested structures into tabular format.
- **Filtering:** Filtered out Falcon 1 launches (retained only Falcon 9 rockets) and launches prior to Nov 13, 2020 (90 launches).
- **Sub-Resource API Calls:** Queried /rockets, /payloads, /launchpads, /cores to retrieve booster version, payload mass, coordinates, and core landing outcome.

4. Data Collection – Web Scraping

Extracting Historical Launch Records with BeautifulSoup

- **Target:** Wikipedia List of Falcon 9 and Falcon Heavy launches (snapshot oldid=1027686922).
- **Parsing:** Extracted tables with class wikitable plainrowheaders collapsible.
- **Data Elements:** Flight number, launch date, booster version, launch site, payload name, payload mass, orbit, customer, launch outcome.
- **Validation:** Scraped dataset cross-verified customer missions and booster versions against API telemetry.

5. Data Wrangling & Feature Engineering

Missing Value Imputation, Class Construction & One-Hot Encoding

- **Missing Values:** PayloadMass missing values imputed with mean (6,104.96 kg). LandingPad nulls kept as explicit category for ocean splashdowns.
- **Target Variable (Class):** Successful landings (True Ocean, True RTLS, True ASDS) = **1 (60 launches, 66.67%)**. Unsuccessful (False Ocean, False RTLS, False ASDS, None None) = **0 (30 launches, 33.33%)**.
- **One-Hot Encoding:** Applied pd.get_dummies to Orbit (11), LaunchSite (3), LandingPad (5), Serial (53) + 8 numeric features = **80 feature columns**.
- **Standardization:** StandardScaler applied to numeric features for distance-based ML models.

6. EDA with Data Visualization

Visualizing Patterns across Flight Number, Payload Mass, Launch Site & Orbits

- **FlightNumber vs LaunchSite:** Early flights concentrated at CCAFS SLC-40; later flights expanded to KSC LC-39A and VAFB SLC-4E with higher success rates (> flight 30).
- **PayloadMass vs LaunchSite:** VAFB handles polar orbits (< 10,000 kg); heavy payloads (> 10,000 kg) launched from CCAFS/KSC with drone ship recovery.
- **Orbit vs Success:** 100% success for ES-L1, GEO, HEO, SSO. GTO success rate is 51.8% due to high orbital velocity requirements.
- **Yearly Trend:** Success rate increased steadily from 0% (2010–2013) to > 84% (2020) as booster hardware matured (Block 5).

7. Exploratory Data Analysis using SQL

Database Queries & Aggregate Metrics

- **Launch Sites:** CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E (55 launches from CCAFS facilities).
- **Payload Metrics:** Total payload mass = 549,446 kg; Min payload = 310 kg; Max payload = 15,600 kg.
- **Recovery Counts:** 41 successful landings on autonomous spaceport drone ships (ASDS).
- **Date Span:** June 4, 2010 to November 5, 2020.

8. Interactive Visual Analytics – Folium Maps

Geospatial Analysis of Launch Sites & Infrastructure Proximity

- **Coastal Proximity:** All launch sites situated adjacent to coastlines (0.5–1.2 km) for safety over water abort trajectories.
- **Marker Clusters:** Folium MarkerCluster visually highlights dense concentration of successful green markers post-2016.
- **Logistics Proximity:** Sites within 1.2–2.5 km of railways and highways for booster transport; > 15 km from major cities for safety.

9. Interactive Visual Analytics – Plotly Dash

Interactive Dashboard for Multi-Variable Launch Success Exploration

- **Dropdown Selector:** Filter by all launch sites or specific sites.
- **Success Pie Chart:** Displays site-specific success vs failure proportions; KSC LC-39A achieved **76.2% success rate**.
- **Payload Slider & Scatter Plot:** Payload mass (0–10,000 kg) vs landing outcome, color-coded by booster version.
- **Key Insight:** Full Thrust (FT) and Block 5 (B5) boosters demonstrate highest reliability across 2,000–6,000 kg payloads.

10. Machine Learning Classification Pipeline

Supervised Classification & GridSearchCV Optimization

- **Dataset Split:** 90 samples split into 72 training (80%) and 18 test (20%) samples with random_state=2.
- **Standardization:** StandardScaler fitted on train set and transformed test set.

• **Models Tuned with GridSearchCV (10-fold CV):**

1. **Logistic Regression:** Best C = 0.01, penalty = l2, solver = lbfgs (CV score: 84.64%).
2. **Support Vector Machine (SVM):** Best C = 1.0, gamma = 0.0316, kernel = sigmoid (CV score: 84.82%).
3. **Decision Tree:** Best criterion = gini, max_depth = 4, min_samples_split = 10, splitter = best (CV score: **87.62%**).
4. **K-Nearest Neighbors (KNN):** Best n_neighbors = 10, algorithm = auto, p = 1 (CV score: 84.82%).

11. Model Evaluation & Performance Comparison

Cross-Validation vs Test Performance & Confusion Matrix Breakdown

• **Test Accuracy:** All 4 optimized models achieved **83.33% accuracy (15/18 correct)** on the unseen test set.

• **Confusion Matrix Analysis:**

- True Positives (TP) = 12 (correctly predicted successful landings)
- True Negatives (TN) = 3 (correctly predicted failed landings)
- False Positives (FP) = 3 (predicted success for failed flights)
- False Negatives (FN) = 0 (**Zero missed successes -> 100% Recall**)

• **Best Model Selection: Decision Tree Classifier** is selected as the top model due to the highest cross-validation score (87.62%) and clear rule interpretability.

12. Conclusions, Business Impact & Strategic Insights

Strategic Recommendations for Commercial Launch Competitors

- **Actionable Bidding Strategy:** Competitors should bid aggressively on heavy payload GTO missions where SpaceX recovery success is lowest (51.8%) and SpaceX cannot reuse the booster, narrowing their cost advantage.
- **Infrastructure Strategy:** Autonomous drone ship recovery is essential for commercial viability, accounting for 68%+ of all recoveries.
- **Reliability Factor:** Falcon 9 Block 5 reusability has reached industrial maturity with > 84% recovery probability.
- **Future Work:** Incorporate maritime weather telemetry (wave height, wind shear) and gradient boosted ensembles (XGBoost).
